4 **(a)** The lowest possible temperature is zero kelvin (0 K).

(i) State the name of this lowest possible temperature.

..... [1]

(ii) Nitrogen boils at 77 K.

Calculate the boiling point of nitrogen on the Celsius scale.

boiling point = ... °C [2]

(b) The temperature of a fixed mass of gas at constant volume changes from 300 K to 400 K.

State and explain, in terms of particles, the effect on the pressure of the gas.

statement .....

explanation .....

.....

.....

.....

[4]

(c) A sample of gas is at a pressure of 120 kPa. The volume of the gas is doubled at constant temperature.

Calculate the new pressure of the gas.

pressure = ... [2]

[Total: 9]

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